Problem Set 9

Submission:

(Attention: Extended deadline!) Monday, 04/05/2026, until 13:30, HKT (UTC+08:00), to be uploaded on the Canvas course homepage.

Only problems marked with $(*)$ will be graded.

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1. **Covariance and correlation**

   (a) Let $X \sim \mathcal{U}([0,a])$ for some $a > 0$. Determine the covariance Cov$[X, X^2]$ and the correlation coefficient $\rho(X, X^2)$.

   (b) Consider rolling a fair die twice, and let X_1 and X_2 denote the outcome of the die on the first and second round, respectively. We define $S = X_1 + X_2$ (sum) and $M = \max\{X_1, X_2\}$ (highest outcome).

      (i) Write down the joint probability mass function $p_{S,M}$ in a table.

      (ii) Calculate Cov$[S, M]$ and $\rho(S, M)$.

   (c) Let $X, Y \sim Ber(p)$ independent random variables, where $p \in (0,1)$. Show that

   $$\text{Cov}[X+Y, X-Y] = 0,$$

   but $X+Y$ and $X-Y$ are not independent.

2. **$(*)$ Decay of covariances in an autoregressive process** [4 Points]

   Let $\sigma, \alpha \in \mathbb{R}$, $|\alpha| < 1$. We consider random variables X_0 and $(Z_n)_{n\in\mathbb{N}}$, independent and assume that $Z_n \sim \mathcal{N}(0,1)$. We define a so-called *autoregressive process* (also called AR(1)-process) inductively by

   $$X_n = \alpha X_{n-1} + \sigma Z_n, \qquad n \in \mathbb{N}.$$

   Prove the following claims.

   (a) If $X_{n-1} \sim \mathcal{N}(m, v)$, then $X_n \sim \mathcal{N}(\alpha m, \alpha^2 v + \sigma^2)$.

   (b) The distribution $\mathcal{N}(0, \sigma^2(1-\alpha^2)^{-1})$ is *stationary*, meaning that

   $$X_0 \sim \mu \qquad \Rightarrow \qquad X_n \sim \mu \text{ for all } n \in \mathbb{N}.$$

   (c) Assuming that $X_0 \sim \mu$, show that

   $$\text{Cov}[X_n, X_{n-k}] = \alpha^k \frac{\sigma^2}{1-\alpha^2}, \qquad 0 \le k \le n.$$

3. **Applications of Jensen's inequality**

   (a) Let $g : \mathbb{R} \to \mathbb{R}$ be convex and $n \ge 1$. Show that for any $\lambda_1, ..., \lambda_n \ge 0$ with $\sum_{j=1}^n \lambda_j = 1$, one has

   $$g\left(\sum_{j=1}^n \lambda_j x_j\right) \le \sum_{j=1}^n \lambda_j g(x_j), \qquad x_1, ..., x_n \in \mathbb{R}.$$

(b) Use (a) and the fact that $\log$ is concave to establish the *arithmetic mean-geometric mean inequality*:

$$\left(\prod_{j=1}^{n} x_j\right)^{\frac{1}{n}} \leq \frac{\sum_{j=1}^{n} x_j}{n}, \qquad x_1, \dots, x_n \geq 0.$$

(c) Let $X \geq 0$ be a non-negative real random variable. Assume that all moments of X exist. Show that

$$\mathbf{E}[X] \leq (\mathbf{E}[X^2])^{\frac{1}{2}} \leq (\mathbf{E}[X^3])^{\frac{1}{3}} \leq \dots$$

4. **($*$) Chernoff bounds reloaded** [4 Points]

In this problem you will prove the following theorem:

> Let $X_1, \dots, X_n$ independent Bernoulli-distributed random variables with $\mathbf{P}[X_i = 1] = p_i$ and $\mathbf{P}[X_i = 0] = 1 - p_i$. Then, one has for $X = \sum_{i=1}^{n} X_i$ and $\mu = \mathbf{E}[X] = \sum_{i=1}^{n} p_i$, and every $\delta > 0$, that $\mathbf{P}[X \geq (1+\delta)\mu] \leq \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu}$.

(a) Prove the previous assertion. Use the following steps:

i. Recall problem 3 on Problem set 7, and apply the result to X.

ii. Replace X by the sum of X_i. Calculate explicitly the expectation appearing in the formula obtained.

iii. Replace $1 + p_i(e^t - 1)$ with an upper bound (via $1 + y \leq e^y$) and simplify.

iv. Find t so that the bound is minimal.

(b) Application: Consider the n-fold toss of a fair coin. Calculate a bound for the probability that the number of "heads" exceeds its expectation by at least 10%, using

i. the Čebyšev-inequality,

ii. the version of the Chernoff bound found above,

if $n = 10^3$ or $n = 10^4$. Compare the results.

Hint: Since $p = 1/2$, the distribution is symmetric.